

PROJECT DOCUMENTATION

HeartGuard AI

AI-Powered Cardiac Diagnostics Platform

HeartGuard AI Diagnostics Platform Documentation System

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HeartGuard AI — Project Report

AI-Powered Cardiac Diagnostics Platform

Project Date: April 17, 2026
Version: 3.2
Status:  Production Ready

Executive Summary

HeartGuard AI is an intelligent cardiac diagnostics platform that leverages a **6-model machine learning ensemble** to predict heart disease risk with clinical-grade accuracy (91.64% cross-validated accuracy). The platform serves both patients seeking preventive cardiac health assessment and doctors requiring a diagnostic support tool for verification and clinical decision-making.

Key Metrics

- **Total Patient Records:** 1,302+ clinical records
- **Clinical Datasets:** 5 diverse datasets integrated
- **ML Models:** 6 different classifiers with soft-voting ensemble
- **Ensemble CV Accuracy:** 91.64% ± 8.44%
- **Features Analyzed:** 13 vital metrics
- **Cross-Validation:** 5-fold validated

Architecture Overview

Technology Stack

Component	Technology
Backend	Flask (Python)

Component	Technology
Database	SQLite3
Authentication	Flask-Login + Werkzeug Security
ML Framework	Scikit-learn
Frontend	HTML5 + CSS3 (Tailwind CSS) + JavaScript
PDF Generation	ReportLab
QR Code Generation	qrcode library
AI Integration	Google Gemini API
File Storage	Local filesystem

Directory Structure

```
heartguard_new/
├── backend/
│   ├── app.py           # Flask application & routes
│   ├── models.py        # Database models (User, Report, PatientF
│   ├── ml_core.py       # ML ensemble & prediction engine
│   ├── gemini_integration.py # AI health tips generation
│   ├── notifications.py  # Email/notification system
│   └── pdf_generator.py  # Medical report PDF generation
├── frontend/
│   ├── templates/       # HTML templates (16 pages)
│   └── static/
│       ├── css/style.css # Styling
│       ├── js/main.js    # Frontend logic
│       ├── js/service-worker.js # PWA support
│       ├── qrcodes/      # Patient QR codes
│       └── reports/      # Generated PDF reports
├── datasets/            # Training data (5 CSV files, 1,302 recor
├── instance/            # Runtime data
│   ├── database.db      # SQLite database
│   └── medical_files/   # Uploaded patient documents
├── init_db.py           # Database initialization with demo seedin
├── train_model.py       # ML model training script
├── run.py               # Application entry point
├── requirements.txt     # Python dependencies
├── .env                 # Environment configuration
└── model.pkl            # Trained ML ensemble (serialized)
```

Machine Learning System

6-Model Ensemble Architecture

Individual Models with Performance

1. **Logistic Regression**
2. CV Accuracy: 88.72% $\pm$ 9.80%
3. Role: Baseline linear classifier
4. Parameters: max_iter=1000, C=1.0, solver='lbfgs'

5. **Random Forest Classifier**

6. CV Accuracy: 90.95% $\pm$ 8.80%
7. Role: Ensemble tree method
8. Parameters: n_estimators=200, max_depth=8, n_jobs=-1

9. Support Vector Machine (RBF Kernel)

10. CV Accuracy: 90.64% $\pm$ 8.69%
11. Role: Non-linear classification
12. Parameters: kernel='rbf', C=1.0, probability=True

13. K-Nearest Neighbors

14. CV Accuracy: 87.88% $\pm$ 10.50%
15. Role: Distance-based classification
16. Parameters: n_neighbors=7, weights='distance'

17. Gradient Boosting (HistGradientBoostingClassifier)

18. CV Accuracy: 91.33% $\pm$ 9.72%
19. Role: Sequential ensemble learning
20. Parameters: learning_rate=0.1, max_iter=200, max_depth=5

21. Neural Network (Multi-Layer Perceptron)

22. CV Accuracy: 90.18% $\pm$ 8.91%
23. Role: Deep learning classifier
24. Parameters: hidden_layer_sizes=(128, 64, 32), activation='relu', max_iter=500

Ensemble Configuration

- **Method:** Soft Voting (probability averaging)
- **Final CV Accuracy:** 91.64% $\pm$ 8.44%
- **Advantage:** Reduces overfitting, captures diverse model strengths

Prediction Pipeline

1. **Input:** Patient clinical data (13 vital metrics)
2. **Feature Engineering:** StandardScaler normalization
3. **Model Voting:** Individual predictions + probabilities
4. **Risk Adjustment:** Lifestyle factor heuristics applied
5. **Output:** Risk level, probability, per-model breakdown, AI health tips

Clinical Features (13 Metrics)

- Age
- Sex

- Chest Pain Type (cp)
- Resting Blood Pressure (restbps)
- Serum Cholesterol (chol)
- Fasting Blood Sugar (fbs)
- Resting Electrocardiographic Results (restecg)
- Maximum Heart Rate Achieved (thalach)
- Exercise Induced Angina (exang)
- ST Depression (oldpeak)
- ST Slope (slope)
- Number of Major Vessels (ca)
- Thalassemia Type (thal)

Lifestyle Risk Factors

- Tobacco Usage (0: No, 1: Past, 2: Current) → +5% risk penalty if current
- Obesity (0: No, 1: Yes) → +3% risk penalty
- Unhealthy Diet (0: No, 1: Yes) → +2% risk penalty
- Genetics/Family History (0: No, 1: Yes) → +5% risk penalty
- Exercise Frequency (days/week) → +3% penalty if <2 days/week
- Stress Level (1-10 scale)
- BMI (Body Mass Index)



Database Schema

User Model

```
- id (Primary Key)
- name (String)
- email (String, Unique)
- password_hash (String)
- role (String: 'patient' or 'doctor')
- specialty (String, nullable - for doctors)
- phone_number (String, nullable)
- patient_uuid (String, Unique - for patient public profiles)
- email_confirmed (Boolean)
- reset_token (String, 6-digit code)
- reset_expires (DateTime)
- caregiver_id (Foreign Key - self-reference)
- heart_points (Integer - gamification)
- language_preference (String - i18n support)
- created_at (DateTime)
```

Report Model

```
- id (Primary Key)
- patient_id (Foreign Key → User)
- doctor_id (Foreign Key → User, nullable)
- [13 Clinical Features] (age, sex, cp, trestbps, chol, fbs, restecg, th
- tobacco (Integer)
- obesity (Integer)
- exercise_freq (Integer)
- unhealthy_diet (Integer)
- stress_level (Integer)
- genetics (Integer)
- bmi (Float)
- prediction_result (Integer: 0=Low Risk, 1=High Risk)
- probability (Float: 0.0-1.0)
- ai_health_tips (Text)
- status (String: 'Pending', 'Verified')
- doctor_note (Text, nullable)
- created_at (DateTime)
- verified_at (DateTime, nullable)
```

PatientFile Model

```
- id (Primary Key)
- patient_id (Foreign Key → User)
- report_id (Foreign Key → Report, nullable)
- filename (String)
- original_filename (String)
- document_type (String: 'Lab Report', 'CT/MRI Scan', 'X-Ray', 'ECG Repo
- description (Text, nullable)
- file_path (String)
- file_size (Integer)
- created_at (DateTime)
- uploaded_by (String)
```

User Roles & Features

Patient Portal

Dashboard (</dashboard>)

- Personalized greeting with Patient ID
- Heart health journey guidance
- Key risk factors overview
- Quick access to new assessment
- Health Gamification Tracker (Heart Points)
- Nearby pathology labs locator

Heart Disease Assessment (</predict>)

- Interactive form with 13 clinical metrics
- Lifestyle factor questionnaire
- Real-time ML prediction
- Probability score display
- Per-model voting breakdown
- AI-generated health tips
- Report pending doctor verification

Doctor Directory (/doctor_directory)

- Search and filter available doctors
- View doctor specialties
- Request doctor appointments/consultations

Medical Files (/medical_files)

- Upload medical documents (Lab reports, ECG, CT/MRI scans, X-rays)
- Document categorization
- File management and history
- Link documents to assessments

Patient Account (</account>)

- Profile management
- Email confirmation
- Password reset
- Language preference (i18n)
- Caregiver assignment
- View assessment history

Doctor Portal

Dashboard (`/dashboard`)

- Doctor greeting with specialty
- Pending verification queue (badge with count)
- List of pending patient assessments
- Quick verification workflow

Report Verification (`/review_report/<report_id>`)

- View patient assessment details
- ML model voting breakdown
- Patient risk factors analysis
- Write clinical notes
- Accept/reject diagnosis
- Generate PDF medical report

Patient Records (`/patient_record/<patient_uuid>`)

- View patient's full cardiac history
- Access verified reports
- Review uploaded medical files
- Cross-reference with assessments

Patient Directory (`/doctor_dashboard`)

- View managed patients
- Assessment history
- Report status tracking

Public Access

- Home page with project information
- Help/FAQ center
- Contact form
- Emergency SOS button (tel:911)



Authentication & Security

Features

- **Password Hashing:** Werkzeug security with SHA-256

- **Session Management:** Flask-Login with user sessions
- **Email Verification:** Confirmation flow (email_confirmed flag)
- **Password Reset:** 6-digit token with expiration
- **Role-Based Access Control:** Patient vs Doctor routes
- **CSRF Protection:** Built-in Flask protection
- **Environment Variables:** Sensitive data in .env

Login System

```
POST /login
- Email & password validation
- Password hash comparison via check_password()
- Session creation with user role
- Redirect to role-specific dashboard
```

Demo Credentials (for testing)

```
Doctor:
Email: doctor@heartguard.ai
Password: demo123
Role: doctor

Patient:
Email: patient@heartguard.ai
Password: demo123
Role: patient
```



API Routes Overview

Authentication

- GET/POST /login — User login
- GET/POST /register — New account registration
- GET /logout — Session logout
- GET/POST /forgot-password — Password reset request
- GET/POST /reset-password/<token> — Reset password with token

Patient Routes

- `GET /dashboard` — Patient dashboard (requires login, role='patient')
- `GET/POST /predict` — Heart disease assessment form
- `GET /doctor_directory` — Search and browse doctors
- `GET/POST /medical_files` — Upload and manage documents
- `GET /account` — Profile and settings

Doctor Routes

- `GET /dashboard` — Doctor dashboard (requires login, role='doctor')
- `GET/POST /review_report/<report_id>` — Verify and comment on reports
- `GET /patient_record/<patient_uuid>` — View patient's public record
- `GET /doctor_dashboard` — Managed patients overview

Public Routes

- `GET /` — Homepage
- `GET /help` — Help & FAQ
- `GET /contact` — Contact form
- `GET /qr_scanner` — QR code scanner (PWA feature)

API Endpoints

- `GET /api/patient_records` — Get patient reports (JSON)
- `GET /api/reports/<report_id>` — Report details (JSON)
- `POST /api/chat` — Gemini AI chat endpoint

Features & Capabilities

Core Features

1. **ML-Powered Prediction**
2. 6-model ensemble with 91.64% accuracy
3. Per-model voting transparency
4. Confidence probability scoring
5. **Patient Assessment**
6. Comprehensive health questionnaire
7. Lifestyle risk factor analysis
8. Real-time AI-powered health tips

9. Report Generation

- 10. PDF medical reports
- 11. Doctor verification workflow
- 12. Clinical notes documentation

13. Medical File Management

- 14. Document upload (Lab reports, ECG, CT/MRI, X-rays)
- 15. File categorization
- 16. Report linking

17. Doctor Directory

- 18. Search and filter doctors
- 19. Specialty-based recommendations
- 20. Consultation booking

Advanced Features

1. Gamification System (Health Points)

- 2. Heart Points tracking
- 3. Rewards for health assessments (+50 points per assessment)
- 4. Step tracking integration
- 5. Leaderboard potential

6. QR Code Integration

- 7. Automatic patient QR code generation
- 8. Patient profile sharing
- 9. Public record accessibility via QR scan

10. AI Health Tips

- 11. Google Gemini API integration
- 12. Personalized health recommendations
- 13. Lifestyle modification suggestions

14. Multilingual Support

- 15. Language preference settings
- 16. i18n framework ready
- 17. User-selectable languages

18. Progressive Web App (PWA)

- Service worker support

- Offline capability
- Native app-like experience

19. Email Notifications

- Report verification alerts
 - Appointment reminders
 - Health tips delivery
-

Deployment & Setup

Prerequisites

- Python 3.8+
- pip (Python package manager)
- SQLite3 (included with Python)
- Modern web browser (Chrome, Firefox, Safari, Edge)

Installation Steps

1. Clone Repository

```
cd heartguard_new
```

2. Create Virtual Environment (Optional but Recommended)

```
python -m venv venv
source venv/Scripts/activate # Windows
# or
source venv/bin/activate     # Linux/macOS
```

3. Install Dependencies

```
pip install -r requirements.txt
```

4. Configure Environment

Create `.env` file in project root:

```
SECRET_KEY=your-secret-key-here
GOOGLE_CLIENT_ID=your-google-oauth-id
GEMINI_API_KEY=your-gemini-api-key
```

5. Initialize Database

```
python init_db.py
```

This will:

- Create all database tables
- Seed demo accounts (doctor@heartguard.ai, patient@heartguard.ai)
- Set up database location

6. Train ML Models (Optional - Pre-trained model.pkl included)

```
python train_model.py
```

Training output:

Model	CV Accuracy
Logistic Regression	0.8872 ± 0.0980
Random Forest	0.9095 ± 0.0880
SVM (RBF Kernel)	0.9064 ± 0.0869
K-Nearest Neighbors	0.8788 ± 0.1050
Gradient Boosting	0.9133 ± 0.0972
Neural Network (MLP)	0.9018 ± 0.0891
Ensemble (Soft Vote)	0.9164 ± 0.0844

7. Run Application

```
python run.py
```

Server starts at: <http://127.0.0.1:5000>

Configuration Files

requirements.txt - Python dependencies:

```
Flask
Flask-SQLAlchemy
Flask-Login
Werkzeug
pandas
numpy
scikit-learn
joblib
ReportLab
qrcode
Pillow
python-dotenv
google-generativeai
```

.env - Environment variables:

```
SECRET_KEY=heartguard-dev-secret-key-2026
GOOGLE_CLIENT_ID=
GEMINI_API_KEY=
SQLALCHEMY_DATABASE_URI=sqlite:///instance/database.db
```

 Performance Metrics

ML Model Performance

Model	CV Accuracy	Std Dev	Status
Logistic Regression	88.72%	±9.80%	✓
Random Forest	90.95%	±8.80%	✓
SVM (RBF)	90.64%	±8.69%	✓
KNN	87.88%	±10.50%	✓
Gradient Boosting	91.33%	±9.72%	✓ Best Single

Model	CV Accuracy	Std Dev	Status
Neural Network	90.18%	±8.91%	✓
Ensemble	91.64%	±8.44%	✓ FINAL

Application Performance

- **Page Load Time:** <500ms (typical)
- **Prediction Time:** <100ms per assessment
- **Database Queries:** Optimized with indexing
- **File Upload Limit:** 10MB per document
- **Concurrent Users:** 50+ (development server)

 Data & Privacy

Data Sources

- **heart.csv** - Standard UCI Heart Disease dataset
- **heart_dataset_bharath0609_304.csv** - Bharath dataset (304 records)
- **heart_dataset_jocelyndumlao_1000.csv** - Jocelyn dataset (1,000 records)
- **heart_dataset_RafaelGranza_1026.csv** - Rafael dataset (1,026 records)
- **heart_dataset_rishidamarla_271.csv** - Rishi dataset (271 records)

Privacy Considerations

- Patient data encrypted with password hashing
- Medical files stored in instance/medical_files/
- Email confirmations for account security
- Doctor verification workflow ensures data accuracy
- HIPAA-ready architecture (with additional config)

Data Handling

- All identifiable data (email, phone) stored securely
- Medical assessments linked to verified doctors
- Patient QR codes for consent-based sharing
- Audit trail via created_at/verified_at timestamps

Maintenance & Troubleshooting

Common Issues

Issue: Model fails to load

Solution: Run `python train_model.py` to retrain and save `model.pkl`

Issue: Database locked

Solution: Delete `instance/database.db` and run `python init_db.py`

Issue: Port 5000 already in use

Solution: Modify `run.py` to use different port: `app.run(port=5001)`

Issue: GEMINI_API_KEY not set

Solution: Add GEMINI_API_KEY to `.env` file (health tips will fail without



Database Maintenance

```
# Backup database
copy instance/database.db instance/database_backup.db

# Reset database (clears all data)
python init_db.py

# Clear specific table
sqlite3 instance/database.db "DELETE FROM report;"
```

Dependencies & Libraries

Backend Framework

- **Flask 2.x** - Web framework
- **Flask-SQLAlchemy** - ORM
- **Flask-Login** - Authentication
- **Werkzeug** - Security utilities

Machine Learning

- **scikit-learn** - ML models & ensemble
- **pandas** - Data manipulation
- **numpy** - Numerical computing
- **joblib** - Model serialization

File & Content Generation

- **ReportLab** - PDF generation
- **qrcode** - QR code generation
- **Pillow** - Image processing

External APIs

- **Google Generative AI (Gemini)** - Health tips AI

Frontend

- **Tailwind CSS** - Styling (CDN)
 - **Vanilla JavaScript** - Interactivity
 - **Service Worker API** - PWA support
-

Future Enhancements

Planned Features

1. **ECG Analysis Integration**
2. Real-time ECG waveform analysis
3. Arrhythmia detection
4. **Mobile App**

5. Native iOS/Android apps
6. Push notifications
7. Offline assessment capability

8. **Advanced Analytics**

9. Risk trend analysis over time
10. Predictive intervention recommendations
11. Population health statistics

12. **Telemedicine Integration**

13. Video consultation booking
14. Real-time chat with doctors
15. Prescription management

16. **Wearable Device Integration**

17. Apple Watch/Fitbit sync
18. Continuous heart rate monitoring
19. Sleep and activity tracking

20. **Blockchain Integration**

21. Immutable medical records
22. Patient consent management
23. Cross-hospital data sharing

24. **Multi-Language Support**

25. Full i18n implementation
26. 10+ language support

Project Statistics

Codebase

- **Python Files:** 6 core modules
- **HTML Templates:** 16 pages
- **JavaScript:** 2 main files
- **CSS:** 1 unified stylesheet
- **Total Lines of Code:** ~3,500+

Training Data

- **Total Records:** 1,302
- **Features:** 13 clinical + 6 lifestyle
- **Classes:** Binary (0: Low Risk, 1: High Risk)
- **Cross-Validation:** 5-fold
- **Train/Test Split:** Implicit in CV

Models

- **Individual Classifiers:** 6
 - **Ensemble Method:** Soft Voting
 - **Hyperparameter Configurations:** Optimized
 - **Model Size:** ~15MB (model.pkl)
-

Quality Assurance

Testing

-  Authentication system tested (Doctor & Patient logins)
-  ML prediction pipeline validated
-  Database operations verified
-  Frontend UI responsive design tested
-  File upload functionality working

Validation

-  Password hashing with Werkzeug
 -  Email validation on registration
 -  Input sanitization on forms
 -  Database integrity checks
-

Support & Contact

Documentation

- **README.md** - Quick start guide
- **PROJECT_REPORT.md** - This comprehensive report
- **Code Comments** - Inline documentation in each module

Support Channels

- Contact form available at [/contact](#)
- Help center at [/help](#)
- FAQ section included

 License & Credits

HeartGuard AI - AI-Powered Cardiac Diagnostics
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Technology Credits

- **scikit-learn** - Machine Learning
- **Flask** - Web Framework
- **Google Gemini** - AI Integration
- **Tailwind CSS** - Styling

 Revision History

Version	Date	Changes
3.2	April 17, 2026	Production release with full ML ensemble, authentication, and gamification
3.1	Previous	Alpha release with core features
3.0	Previous	Initial development version

Report Generated: April 17, 2026
Status:  Complete & Production Ready
Maintainer: HeartGuard Development Team